

Annex 162B

(normative)

Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M

162B.1 Test fixtures

Transmitter and receiver measurements at TP2 or TP3 for the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 hosts (see Annex 162D) and at TP1a or TP4a for the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M hosts (see Annex 120G), are made utilizing the test fixture specified in 162B.2. Cable assembly measurements for the cable assembly types (see Annex 162D) are made between TP1 and TP4 with test fixtures as specified in 162B.3 on both ends. The test fixtures are specified in a mated state to enable connections to measurement equipment. The reference insertion loss of the mated test fixtures is 6.6 dB at 26.56 GHz using Equation (162B–5). The requirements in the referenced subclauses are not the MDI connector specifications for an implemented design.

162B.2 TP2 or TP3 test fixture

The TP2 or TP3 test fixture (also known as Host Compliance Board) is required for measuring the transmitter and receiver specifications at TP2 and TP3. The TP2 and TP3 test points are illustrated in Figure 162–2.

162B.2.1 TP2 or TP3 test fixture insertion loss

The TP2 or TP3 test fixture printed circuit board (PCB) insertion loss values determined using Equation (162B–1) shall be used as the TP2 or TP3 test fixture reference insertion loss. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{tfref}(f) = 1.02(0.001 + 0.24\sqrt{f} + 0.046f) \quad (162B-1)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{tfref}(f)$ is the test fixture PCB reference insertion loss in dB at frequency f
 f is the frequency in GHz

The TP2 or TP3 test fixture PCB reference insertion loss is illustrated in Figure 162B–1.

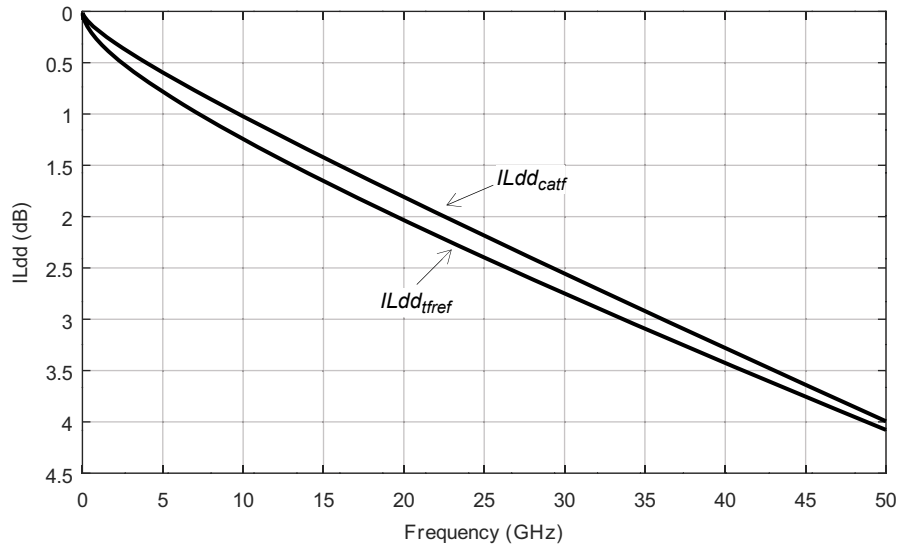


Figure 162B-1—Test fixtures PCB reference insertion losses

162B.3 Cable assembly test fixture

The cable assembly test fixture (also known as Module Compliance Board) is required for measuring the cable assembly specifications in 162.11 at TP1 and TP4. The TP1 and TP4 test points are illustrated in Figure 162-2.

162B.3.1 Cable assembly test fixture insertion loss

The cable assembly test fixture PCB and test point insertion loss values determined using Equation (162B-2) shall be used as the test fixture reference insertion loss. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{catf}(f) = 1.073(-0.00125 + 0.12\sqrt{f} + 0.0575f) \quad (162B-2)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{catf}(f)$ is the test fixture PCB reference insertion loss in dB at frequency f
 f is the frequency in GHz

The cable assembly test fixture reference insertion loss is illustrated in Figure 162B-1.

162B.4 Mated test fixtures

The TP2 or TP3 test fixture and the cable assembly test fixture are specified in a mated state illustrated in Figure 92–18. The mated test fixtures specifications are given below.

162B.4.1 Mated test fixtures insertion loss

The insertion loss of the mated test fixtures shall meet the values determined using Equation (162B–3) and Equation (162B–4).

$$ILdd(f) \leq ILdd_{MTFmax}(f) = \begin{cases} 1.185\sqrt{f} - 0.072f + 0.007f^2 & 0.01 \leq f < 40 \\ 1.915f - 60.78 & 40 \leq f \leq 50 \end{cases} \quad (162B-3)$$

$$ILdd(f) \geq ILdd_{MTFmin}(f) = 1.0225(0.0656\sqrt{f} + 0.164f) \quad (162B-4)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd(f)$	is the measured insertion loss in dB at frequency f
$ILdd_{MTFmax}(f)$	is the maximum mated test fixture insertion loss in dB at frequency f
$ILdd_{MTFmin}(f)$	is the minimum mated test fixture insertion loss in dB at frequency f
f	is the frequency in GHz

The reference insertion loss of the mated test fixtures is determined using Equation (162B–5).

$$ILdd_{MTFref}(f) = 0.8153\sqrt{f} + 0.003405f^2 \quad (162B-5)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{MTFref}(f)$	is the reference insertion loss of the mated test fixture in dB at frequency f
f	is the frequency in GHz

The mated test fixtures insertion loss limits are illustrated in Figure 162B–2.

The FOM_{ILD} is calculated according to 93A.4 with $f_b=53.125 \text{ GHz}$, $T_i=8.5 \text{ ps}$, and $f_r=0.75 \times f_b$. The fitted insertion loss and insertion loss deviation are computed over the range $f_{min}=0.05 \text{ GHz}$ to $f_{max}=40 \text{ GHz}$. FOM_{ILD} shall be less than or equal to 0.15 dB.

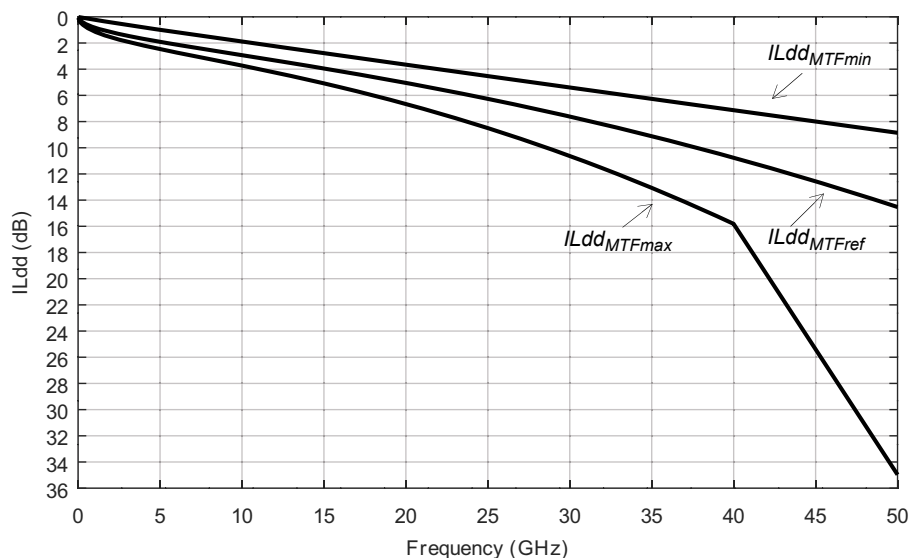


Figure 162B-2—Mated test fixtures insertion loss

162B.4.2 Mated test fixtures effective return loss (ERL)

The values of the mated test fixtures ERL are computed using the procedure in 93A.5 with the parameter values in Table 162B-1. Parameters that do not appear in Table 162B-1 take values from Table 162-20.

The mated test fixture ERL shall be greater than or equal to 10.3 dB.

Table 162B-1—Mated test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—
Target detector error ratio	DER_0	10^{-5}	—
NOTE—The mated test fixture test connector and transmission line are not time-gated (by setting T_{fx} to 0) in order to include the entire test fixture.			

162B.4.3 Mated test fixtures common-mode to differential-mode insertion loss

The common-mode to differential-mode insertion loss of the mated test fixtures measured in both directions shall meet the values determined using Equation (162B–6).

$$IL_{dc}(f) \geq \begin{cases} 30 - (21/28)f & 0.01 \leq f < 20 \\ 15 & 20 \leq f \leq 50 \end{cases} \quad (162B-6)$$

where

$IL_{dc}(f)$ is the common-mode to differential-mode insertion loss in dB at frequency f
 f is the frequency in GHz

The mated test fixtures common-mode to differential-mode insertion loss limit is illustrated in Figure 162B–3.

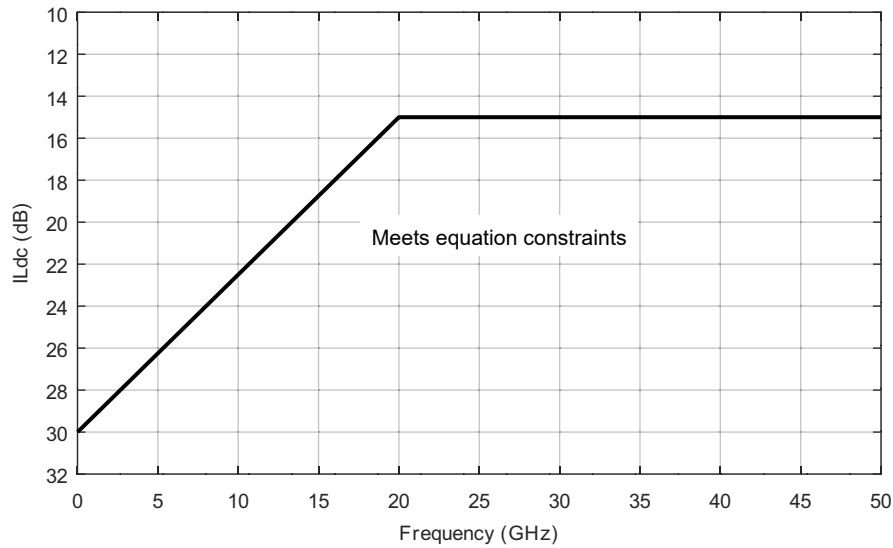


Figure 162B–3—Mated test fixtures common-mode to differential-mode insertion loss

162B.4.4 Mated test fixtures common-mode to common-mode return loss

The common-mode to common-mode return loss of the mated test fixtures measured at each test fixture test interface shall meet the values determined using Equation (162B–7).

$$RL_{cc}(f) \geq \begin{cases} 12 - 9f & 0.01 \leq f < 1 \\ 3 & 1 \leq f < 4 \\ 2.6 + 0.1f & 4 \leq f < 30 \\ 9.5 - 0.13f & 30 \leq f \leq 50 \end{cases} \quad (162B-7)$$

where

$RL_{cc}(f)$ is the common-mode to common-mode return loss in dB at frequency f
 f is the frequency in GHz

The mated test fixtures common-mode to common-mode return loss limit is illustrated in Figure 162B–4.

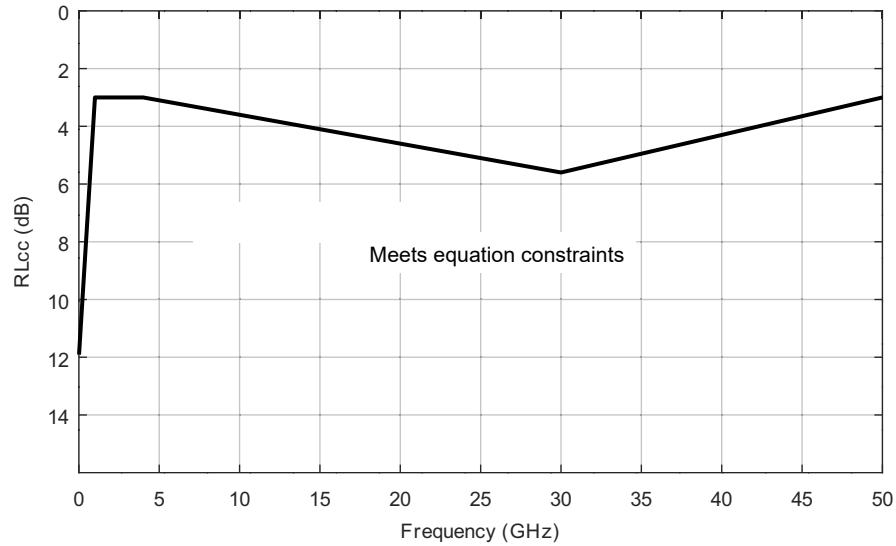


Figure 162B–4—Mated test fixtures common-mode to common-mode return loss

162B.4.5 Mated test fixtures common-mode to differential-mode return loss

The common-mode to differential-mode return loss of the mated test fixtures measured at each test fixture test interface shall meet the values determined using Equation (162B–8).

$$RL_{dc}(f) \geq \begin{cases} 30 - (30/25.78)f & 0.01 \leq f < 12.89 \\ 17.85 - 0.225f & 12.89 \leq f < 35 \\ 10 & 35 \leq f \leq 50 \end{cases} \quad (162B-8)$$

where

$RL_{dc}(f)$ is the common-mode to differential-mode return loss in dB at frequency f
 f is the frequency in GHz

The mated test fixtures common-mode to differential-mode return loss limit is illustrated in Figure 162B–5.

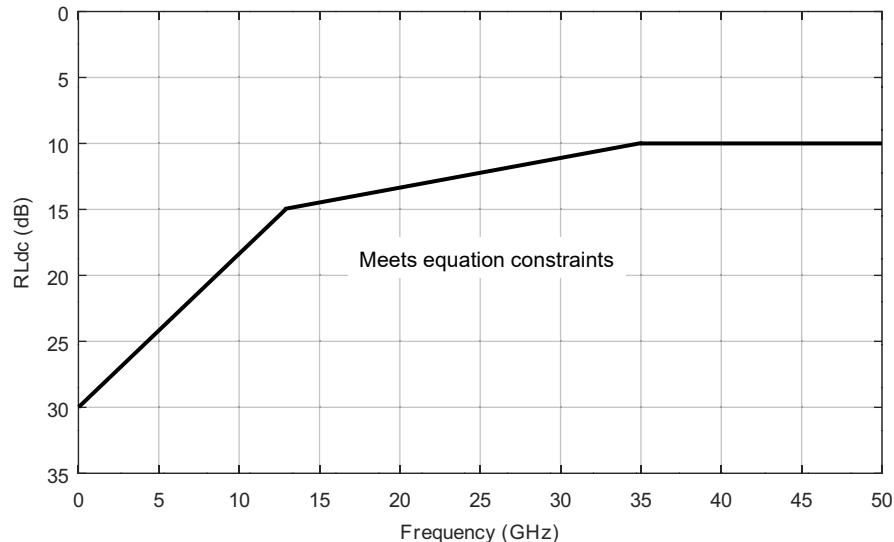


Figure 162B-5—Mated test fixtures common-mode to differential-mode return loss

162B.4.6 Mated test fixtures integrated crosstalk noise

The SFP112 mated test fixture integrated near-end crosstalk noise voltage for the disturber near-end crosstalk loss is determined according to the method in 110B.1.3.6, given the disturber near-end crosstalk loss $NEXT_loss(f)$ is measured over N uniformly spaced frequencies f_n spanning the frequency range 50 MHz to 40 000 MHz with maximum frequency spacing of 10 MHz, and the parameters shown in Table 162B-2. The mated test fixture integrated near-end crosstalk noise voltage shall meet the specification in Table 162B-3.

Table 162B-2—SFP112 mated test fixture integrated near-end crosstalk noise parameters

Description	Symbol	Value	Units
Symbol rate	f_b	53.125	GBd
3 dB reference receiver bandwidth	f_r	39.84	GHz
Near-end disturber peak differential output amplitude	A_{nt}	600	mV
Near-end disturber 20% to 80% rise and fall times	T_{nt}	8.5	ps

Table 162B-3—SFP112 mated test fixture integrated near-end crosstalk noise voltage

Parameter	Value	Units
Integrated near-end crosstalk noise voltage	Less than 1.6	mV

The multi-lane mated test fixtures integrated crosstalk noise voltages are determined using Equation (92–44) through Equation (92–48), given the multiple disturber near-end crosstalk loss $MDNEXT_loss(f)$ and multiple disturber far-end crosstalk loss $MDFEXT_loss(f)$ measured over N uniformly-spaced frequencies f_n spanning the frequency range 50 MHz to 40 000 MHz with a maximum frequency spacing of 10 MHz, and the parameters shown in Table 162B–4. The multi-lane mated test fixture integrated crosstalk noise shall meet the specifications in Table 162B–5.

Table 162B–4—Multi-lane mated test fixture integrated crosstalk noise parameters

Description	Symbol	Value	Units
Symbol rate	f_b	53.125	GBd
3 dB reference receiver bandwidth	f_r	39.84	GHz
Near-end disturber peak differential output amplitude	A_{nt}	600	mV
Far-end disturber peak differential output amplitude	A_{ft}	600	mV
Near-end disturber 20% to 80% rise and fall times	T_{nt}	8.5	ps
Far-end disturber 20% to 80% rise and fall times	T_{ft}	8.5	ps

Table 162B–5—Multi-lane mated test fixture integrated crosstalk noise

Parameters	Value	Units
MDFEXT integrated crosstalk noise voltage	less than 4.2	mV
MDNEXT integrated crosstalk noise voltage	less than 1.5	mV
Total integrated crosstalk noise voltage	less than 4.4	mV

162B.5 Protocol implementation conformance statement (PICS) proforma for Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M²⁶

162B.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

162B.5.2 Identification

162B.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

162B.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

²⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

162B.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
HTF1	TP2 or TP3 test fixture	162B.2		O	Yes [] No []
CATF1	Cable assembly test fixture	162B.3		O	Yes [] No []

162B.5.4 PICS proforma tables for test fixtures for Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Mated test fixtures insertion loss	162B.4.1	Per Equation (162B–3) and Equation (162B–4)	M	Yes []
TF2	Mated test fixtures FOM_{ILD}	162B.4.1	Less than or equal to 0.15 dB	M	Yes []
TF3	Mated test fixtures ERL	162B.4.2	Greater than or equal to 10.3 dB	M	Yes []
TF4	Mated test fixtures common-mode to differential-mode insertion loss	162B.4.3	Per Equation (162B–6)	M	Yes []
TF5	Mated test fixtures common-mode to common-mode return loss	162B.4.4	Per Equation (162B–7)	M	Yes []
TF6	Mated test fixtures common-mode to differential-mode return loss	162B.4.5	Per Equation (162B–8)	M	Yes []
TF7	SFP112 Mated test fixtures integrated near-end crosstalk noise	162B.4.6	Meets the requirements of Table 162B–3	M	Yes []
TF8	Multi-lane mated test fixtures integrated crosstalk noise	162B.4.6	Meets the requirements of Table 162B–5	M	Yes []